

COLLECTIVE
MINDS

SMART INDIA HACKATHON 2026

ROUTE HAZARD AI

- ▶ Problem Statement ID - SIH26001
- ▶ Problem statement Title - **Smart Urban Road safety System for & Automated Manhole Hazard Mitigation system**
- ▶ Theme-Disaster Management
- ▶ PS Category - Software
- ▶ Team Name - Collective Minds



Proposed Solution

- **Detailed Explanation:**

1. **Sensor less Hazard Mapping:** An offline – first mobile app navigation system that alerts drivers to open manholes ,high water levels , and toxic gas risks without relying on expensive hardware sensors inside drains.
2. **Multi-source Data Engine:** Combines past municipal maintenance records . Offline crowdsourced user reports to pinpoint road hazards accurately.

- **How it Addresses the Problem:**

1. **Hands-Free Proactive Safety:** delivers instant, offline 100 meter voice alerts to commuters ahead of hazard spots, preventing night time accidents.
2. **Automated Municipal Workflow:** instantly queues and syncs geotagged repair alerts to civic bodies when internet connectivity is restored ,streamlining repair dispatches.

- **Innovation & Uniqueness:**

1.Fail-Safe Mechanical Net: : A physical, non-electronic safety net installed underneath the manhole cover that catches debris, pedestrians, or two-wheeler riders even if human repair is delayed or the lid is missing .

2.100% Offline-First Navigation: Operates completely without internet connectivity during severe monsoons and network outages by leveraging locally cached danger maps and device-to-device data sync.



TECHNOLOGIES TO BE USED

- Python – AI/ML development
- YOLO + Computer vision – hazard detection
- Open CV – Image/Video processing
- HTML , CSS , JavaScript – Dashboard
- GPS + Maps/GIS – Location Mapping
- Database – Hazard records

HAZARDS DETECTED

- Potholes
- Road Cracks
- Water Logging
- Road obstacles
- Damage/Missing road signs
- Other visible road hazards

• Implementation Flow



WHY THE IDEA IS FEASIBLE

- Uses existing smartphones
- AI detects visible hazards from images/video
- GPS provides location tagging
- Web dashboard supports authority monitoring
- Uses accessible technologies and open-source frameworks

CHALLENGES&RISKS

- Poor lightning and night conditions
- Rain , glare and waterlogging
- False detection
- GPS inaccuracies
- Duplicate reports
- large video/data volume

STRATEGIES

- Use diverse day/night/weather datasets
- confidence threshold + frame verification
- Repeated GPS detections for validation
- offline storage + later synchronization
- process selected frames to reduce data load

SCABILITY ROADMAP

1.Prtotype
Mobile camera+AI



2.Field Test
Selected roads



3.Smart Map
Live hazard map



4.Authority
Prioritized reports



5.Scale
City-wide
monitoring

EARLIER DETECTION → FASTER REPORTING → BETTER MAINTAINANCE → SAFER ROADS

POTENTIAL IMPACT



Finds problems early – Detects high water levels, open covers, and dangerous gases.



Prevents accidents – Helps keep people safe from unsafe manholes.



Protects workers – Warns workers about dangerous conditions.



Quick action – Sends alerts to authorities when there is a problem.



Safer city – Helps make roads and drainage systems safer.

BENEFITS



SOCIAL BENEFITS

- Keeps people safe.
- Reduces accidents on roads.



ECONOMIC BENEFITS

- Saves time.
- Reduces unnecessary maintenance work.



ENVIRONMENTAL BENEFITS

- Helps prevent drainage overflow.
- Keeps roads and surroundings cleaner.

MAIN IMPACT



Detect the problem early



Send an alert



Take quick action



Keep the city safe

RESEARCH & REFERENCES

- ▶ Ministry of Road Transport & Highways(MORTH):Annual accident reports highlighting urban road infrastructure hazards and open manhole fatalities.
- ▶ IEEE Standards (Smart Cities): Frameworks on crowdsourced urban infrastructure monitoring and participatory sensing architectures
- ▶ Swachh Bharat & Digital India Mission: Guidelines for leveraging mobile governance in grievance redressal.
- ▶ Computer Vision in Civil Infrastructure: Research papers on automated pavement and asset anomaly detection using deep learning (CNNs)
- ▶ GIS & Spatial Indexing: PostGIS spatial database implementation for high-density urban hazard cluster mapping.
- ▶ Open Street Map Foundation: Geolocation precision standards and offline-first mobile caching methodologies
- ▶ Municipal SLA Compliance Protocols: Standard operating procedures for automated ticket routing and escalation trees in urban bodies

App Usage & Stakeholder Workflow

- For Citizens : Simple 3-click reporting process:

Open App > Snap Hazard Photo > Auto-Submit with GPS.

Track repair updates transparently.

- For Field Workers : Receive push notifications for critical repairs in assigned zones with exact map navigation to resolve and upload closure proofs.
- For Municipal Admin : Analyze city-wide hazard heatmaps, monitor contractor response times, and allocate budgets based on infrastructural vulnerability.